Risk-certified class expansion through reference-anchored evidence alignment

Junyi Zhang[a,*]

[a]*Shandong University, China*

Abstract

Class expansion must integrate few-shot novel evidence while preserving base-class accuracy. We couple vocabulary-invariant reference alignment with finite-sample admission certification. Fixed references preserve existing rankings; query normalization induces an adaptive admission margin. Preserved base rankings make protection equivalent to the retention requirement, while a direct test credits replacement readouts' recoveries. On ImageNet-R, independently tuned query normalization adds 3.60 and 5.29 percentage points of novel capacity over uncentered fusion at 95% retention with L/14 and B/32. In the original frozen L/14 readouts, retained and dense capacities are comparable, yet directly certified policies differ by 4.85 points. Separately, original source-selected models reach 86.82% and 58.96% novel accuracy on ImageNet-Sketch with target calibration and empirical retention attainment in all ten partitions per encoder.

Keywords: Class expansion, Retention constraints, Incremental learning, Evidence fusion, Vision–language models, Long-tailed recognition

*Corresponding author.
Email address: cheerssom0876@gmail.com (Junyi Zhang)

1. Introduction

An expanding recognition system integrates unequal evidence: a trained classifier encodes accumulated base supervision, whereas arriving classes provide names and a few examples. Deployment requires novel admission without exceeding a specified loss of base recognition. Evidence must remain comparable as the vocabulary grows, and finite calibration data must support the resulting retention requirement.

Vocabulary growth can disturb evidence alignment even with frozen models. Standardizing a branch against active candidates changes its scale when a class arrives; after fusion, existing classes can exchange rank. Fixed base reference banks remove this dependence. Their query-specific statistics adapt learned, semantic and prototype evidence, while a scalar offset controls novel admission.

We develop *risk-certified class expansion through reference-anchored evidence alignment*. Preserving the original base ranking makes every lost correct base prediction go to a novel class and prevents recovery of base errors. This structure makes a sufficient protection criterion exactly match the retention requirement. A second, direct accuracy test credits replacement readouts' recoveries under the same requirement. The framework exposes how recognition capacity becomes useful certified admission.

The contributions are:

- A vocabulary-invariant alignment of asymmetric evidence, with an explicit query-adaptive admission margin and an exact retention envelope for fixed readouts.

- Finite-sample admission control for class-macro accuracy over prespecified arrivals, including a tail-coverage condition for the protection certificate and a direct test that credits recovered reference errors.

- A capacity-to-policy analysis connecting error structure, attainable recognition and selected admission. Controlled fusion comparisons establish the alignment mechanism; source-frozen ImageNet-Sketch evaluation demonstrates target calibration without model retraining.

On ImageNet-R L/14, retained and dense readouts have comparable capacity at the minimum retention target, while their directly certified policies differ by 4.85 points. The analysis therefore concerns both recognition and its use under an accuracy requirement.

2. Related work

Long-tailed recognition addresses unequal class supervision through balanced fitting and losses [1, 2, 3]. CLIP supplies semantic image–text evidence [4], while prompt adaptation [5, 6], prototypes [7, 8] and semantic caches [9] adapt pre-trained representations to labeled data. Our setting combines these evidence types according to their availability at class arrival.

Class-incremental and few-shot incremental learning study retention across successive tasks [10, 11]. SimpleCIL [12] and RanPAC [13] show the value of pre-trained prototypes and analytic readouts. KBS approaches develop multi-feature similarity supplementation [14], sharpness-aware prototype calibration [15], adaptive exploration [16], representation-enhanced analytic learning [17], and multimodal prompt fusion [18].

Recent work aligns task-specific expert inference in MoTE [19], separates expanding feature spaces through orthogonal constraints [20], and learns invariant representations through counterfactual distribution intervention [21]. We study accuracy-constrained admission with asymmetric base and novel evidence. CLIP-guided discovery [22] additionally uses unlabeled novel data, whereas our arriving classes provide names and labeled support. FeNeC [23] supplies a clustered-representative readout for controlled frozen-feature comparisons.

BiMC [24] and HyCal [25] calibrate multimodal or heterogeneous recognition metrics. Calibrated seen/unseen recognition [26] studies the related operating trade-off, and Learn then Test [27] supplies risk-controlled selection. Our design couples vocabulary-invariant evidence alignment with finite-sample admission control. Query moments shape the recognition frontier; independent calibration selects a policy. The statistical constructions build on risk-controlled testing and bounded-mean betting [28]. We derive their consequences for class-macro retention, unequal calibration coverage and retained versus replacement error structures. An author-code-checked HyCal transfer supplies a matched heterogeneous-metric comparison.

3. Reference alignment and certified admission

3.1. Vocabulary-invariant alignment of asymmetric evidence

Base classes $\mathcal{B}$ have labeled, potentially imbalanced fitting data. Arriving classes $\mathcal{N}$ provide a name and k labeled support images per class. Prediction selects from $\mathcal{B} \cup \mathcal{N}$ without a group oracle. A frozen encoder produces a normalized feature $v = f(x)$; normalized text anchors t_c average eight fixed

prompts. Learned base scores are $q_L(x, b) = v^\top w_b + b_b$, fitted by inverse-frequency-weighted ridge regression. Novel prototypes are normalized support means p_n, giving $q_P(x, n) = v^\top p_n$; semantic scores are $q_T(x, c) = v^\top t_c$.

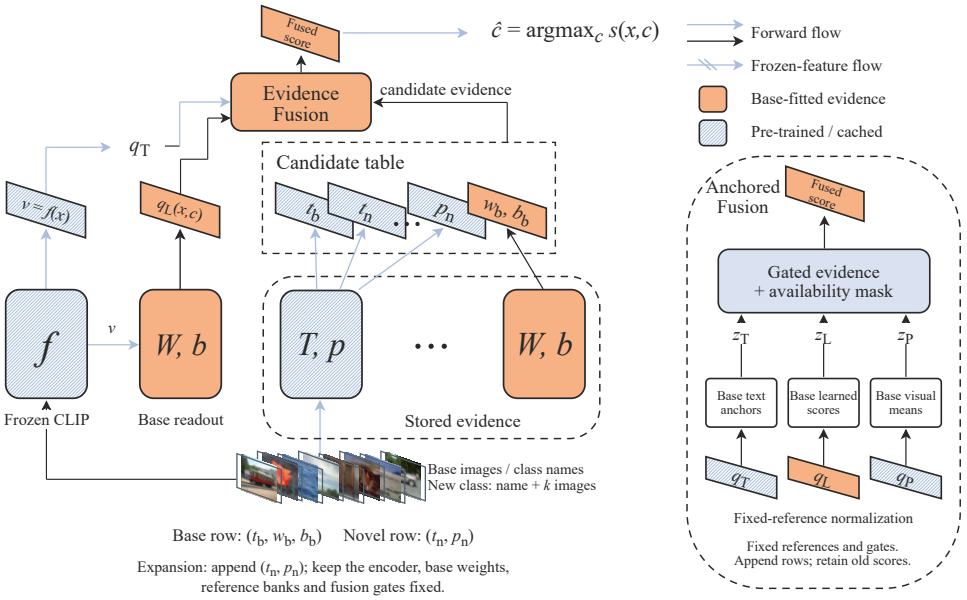


Figure 1: Reference-anchored asymmetric readout. Fixed base references set each branch's score frame; learned base columns remain stored as novel text anchors and support prototypes are appended. The separate novel offset controls admission. Orange marks base-fitted evidence; blue hatching marks pre-trained or cached evidence.

For each evidence source e, a fixed base reference bank $\mathcal{R}_e$ defines

$$\mu_e(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} q_e(x, b), \tag{1}$$

$$\sigma_e^2(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} (q_e(x, b) - \mu_e(x))^2. \tag{2}$$

We use base text anchors, learned columns and visual means as the text,

learned and prototype references. Normalized evidence is

$$\widehat{q}_e(x,c) = \frac{q_e(x,c) - \mu_e(x)}{\max(\sigma_e(x), 10^{-6})}. \tag{3}$$

The statistics adapt to each query while remaining independent of the active novel vocabulary. Fusion uses the evidence available to each row:

$$s(x,b) = \widehat{q}_T(x,b) + a\widehat{q}_L(x,b), \qquad b \in \mathcal{B}, \tag{4}$$

$$s(x,n) = \widehat{q}_T(x,n) + u\widehat{q}_P(x,n) + d, \qquad n \in \mathcal{N}. \tag{5}$$

The coefficients a, u combine evidence; d controls novel admission (Fig. 1). Removing the text-centering shift shared by every candidate exposes the effective offset

$$d_{\text{eff}}(x) = d + a\frac{\mu_L(x)}{\sigma_L(x)} - u\frac{\mu_P(x)}{\sigma_P(x)}, \tag{6}$$

using the floored scales in Eq. (3). Centering adjusts the group margin for each query; scaling also changes within-group fusion.

Why the reference must be fixed. Consider old text scores $(0, 2)$ and a learned contribution $(0.25, -0.25)$. Candidate-standardized text scores $(-1, 1)$ make the second old class win after fusion. Appending a text score of 100 changes the old standardized scores to approximately $(-0.728, -0.685)$, reversing their fused order without changing either old class's parameters. Fixed reference banks prevent this vocabulary-induced change.

Proposition 1 (Vocabulary-growth invariance)**.** *With fixed input, encoder, existing evidence, reference banks, coefficients and tie-breaking, appending classes preserves every existing score and its within-vocabulary ordering.*

Proof. Every term in Eqs. (2)–(5) for an existing class remains fixed. Hence each pairwise score difference remains fixed. □

This property concerns stored discrimination; a newly appended class can still win the expanded prediction. Raw and fixed-temperature readouts also preserve scores, providing controls that isolate query alignment.

Proposition 2 (Ridge identity and adaptive admission margin)**.** *Let the weighted ridge fit have one-hot targets, an unregularized intercept and a unique solution. Its learned reference mean satisfies* $\mu_L(x) = 1/B$*, where* $B = |\mathcal{B}|$*. Consequently,*

$$d_{\text{eff}}(x) = d + \frac{a}{B\sigma_L(x)} - u\frac{\mu_P(x)}{\sigma_P(x)}. \tag{7}$$

Proof. Let X include the constant feature, Y contain one-hot targets, Ω contain sample weights, and L regularize slopes. For the intercept vector e_0, $Xe_0 = \mathbf{1}$, $Le_0 = 0$ and $Y\mathbf{1} = \mathbf{1}$. The normal equations give $(X^\top\Omega X + L)W\mathbf{1} = (X^\top\Omega X + L)e_0$. Uniqueness implies $W\mathbf{1} = e_0$; summing the B learned scores gives one. Substitution into Eq. (6) completes the result. □

Thus the learned part of the adaptive offset varies through score dispersion, while the prototype reference contributes both its query mean and dispersion. The combined margin determines admission. Each new class needs $O(D)$ storage and $O(kD)$ prototype construction after encoding; scoring K classes costs $O(KD)$.

3.2. The retention envelope

A base-only pseudo-expansion episode splits base classes into pseudo-base and pseudo-novel groups. A classifier fitted on pseudo-base classes is combined with names and reserved support for pseudo-novel classes. Held-out base queries select a, u, d by harmonic macro accuracy,

$$H = 2A_B A_N/(A_B + A_N), \tag{8}$$

with zero assigned when both accuracies vanish. The coefficients then remain fixed across actual arrivals. Actual novel query labels and test labels are excluded from this selection; grids and split seeds are in Supplementary Section K.

Let $A_{B,\mathrm{ref}}^{\mathrm{closed}}$ denote the retained readout's closed-base macro accuracy. At retention fraction $0 < r < 1$, every readout receives the same absolute base target: a 90% reference gives an 85.5% target at 95% retention. Empirical expansion capacity is

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\mathrm{ref}}^{\mathrm{closed}}. \tag{9}$$

For a fixed readout, let h_B, h_N be the best labels within each group and $m(x)$ their score difference before adding d. Increasing d switches each query from h_B to h_N once, at $m(x)$, subject to the fixed tie rule. Thus $A_B(d)$ is nonincreasing and $A_N(d)$ is nondecreasing. The largest feasible event maximizes Eq. (9) for retained and replacement readouts. Test-label sweeps describe capacity; independent calibration selects deployable thresholds.

Corollary 1 (Preserved base error structure)**.** *Let R indicate correctness of the fixed retained readout restricted to base classes and C_d correctness after novel admission. Then $C_d \leq R$ on base queries. Consequently recovery $G = \Pr(R = 0, C_d = 1) = 0$, and all destruction $D = \Pr(R = 1, C_d = 0)$ goes to novel labels:*

$$A_B(d) = A_{\mathrm{ref}} - D. \tag{10}$$

The reference in this statement uses the same frozen fusion coefficients. Independently retuned readouts are compared experimentally against the common original reference, relative to which recovery is allowed.

3.3. Finite-sample retention certification

For every readout, let R denote correctness of the same original closed-base reference and C_d correctness after expansion. With destruction D and recovery G, $A_B = A_{\text{ref}} - D + G$. Define the protection loss

$$L_d = R(1 - C_d) + (1 - r)(1 - R), \qquad \alpha = 1 - r. \tag{11}$$

Its risk is $D + \alpha(1 - A_{\text{ref}})$. Thus $\mathbb{E}L_d \leq \alpha$ is sufficient for retention and, when $G = 0$, equivalent to $A_B \geq rA_{\text{ref}}$. This is the direct connection between preserved base rankings and the accuracy-control objective.

We specialize bounded-loss concentration within the Learn then Test framework [27] to class-macro retention. The resulting weighted certificate exposes the role of tail-class coverage.

Theorem 1 (Macro protection and tail coverage)**.** *For independent class-stratified calibration losses in* $[0, 1]$*, set* $w_i = 1/(Bn_{c(i)})$*,* $b = \max_i w_i$*, and* $q = \sum_i w_i L_i$*. A valid p-value for the null* $\sum_i w_i \mathbb{E}L_i \geq \alpha$ *is*

$$p(q) = \begin{cases} \exp[-\operatorname{kl}(q, \alpha)/b], & q < \alpha, \\ 1, & q \geq \alpha. \end{cases} \tag{12}$$

At level δ/T*, passing this bound requires*

$$n_{\text{eff}} = 1/b = B \min_c n_c \geq \frac{\log(\delta/T)}{\log r}. \tag{13}$$

Proof. For $\mu = \sum_i w_i \mathbb{E}L_i$ and $x < \mu$, the bounded-loss Chernoff argument yields $\Pr(q \leq x) \leq \exp[-\operatorname{kl}(x, \mu)/b]$ (Supplementary Section J). Monotonicity in $\mu \geq \alpha$ gives the p-value. Its minimum $p(0) = r^{1/b}$ gives Eq. (13). □

The least-covered class controls this certificate's effective count. The condition is necessary for this bound; observed losses determine whether a covered candidate passes.

We adapt bounded-mean betting [28] to paired correctness with class-macro weights, so that the direct test credits recovered reference errors.

Theorem 2 (Recovery-aware direct accuracy test)**.** *For independent correctness pairs with the same macro weights, let* $Z_i = C_{d,i} - rR_i$*. For any fixed* $0 \le \lambda < 1/r$*,*

$$E_\lambda = \prod_i [1 + \lambda(w_i/b)Z_i] \tag{14}$$

is nonnegative and has expectation at most one under $\sum_i w_i \mathbb{E}Z_i \le 0$*. A fixed convex mixture* E *therefore gives a valid fixed-sample p-value* $\min(1, 1/E)$ *for the unsafe accuracy null.*

Proof. Each factor is positive because $Z_i \ge -r$ and $w_i/b \le 1$. Independence and $1 + x \le e^x$ give $\mathbb{E}E_\lambda \le \exp[(\lambda/b)\sum_i w_i \mathbb{E}Z_i] \le 1$. Linearity and Markov's inequality establish the mixture result. □

We use twelve fixed mixture components, specified in Supplementary Section N. Sample sizes and mixture weights are fixed independently of calibration outcomes.

Both tests operate on thresholds fixed by fitting/tuning data, ordered from conservative to permissive. Fixed-sequence testing [27] stops at the first failure at level δ/T and returns the last accepted threshold. If none passes, the policy uses the original reference with zero novel admission. A union bound controls unsafe acceptance across $T = 5$ prespecified arrivals with probability at most δ. The statement holds per fixed method/order/partition/policy

under independent target class-conditional sampling; calibration data are excluded from fitting and tuning. The reference fallback and finite admission are reported separately.

Error structure and paired-score variability. For a class-macro base query, let $A_C = \mathbb{E}C$ and $A_R = \mathbb{E}R$. Direct expansion of $Z = C - rR$ gives

$$\begin{aligned} \mathrm{Var}(Z) = A_R(1-r)^2 + (A_R - A_C)(2r-1) \\ + 2rG - (A_C - rA_R)^2. \end{aligned} \tag{15}$$

At fixed A_R, A_C, recovery accompanied by additional destruction increases variance by $2rG$ relative to a feasible zero-recovery coupling. Retention removes that component while also making protection exact. For stratified calibration, the variance of the weighted mean is $\sum_c \mathrm{Var}(Z \mid c)/(B^2 n_c)$. The analogous variance comparison requires matching both accuracies within every class and sharing calibration counts. Actual certification also depends on accuracy slack, class coverage and the threshold family. Supplementary Section P derives the identity and its scope.

4. Experiments

4.1. Protocol and comparisons

We evaluate CIFAR-100 [29], Oxford-IIIT Pets [30] and ImageNet-R [31], using frozen OpenAI CLIP ViT-L/14 and ViT-B/32 features, five-shot support, ten frequency-stratified class partitions and two five-session arrival orders. Base/novel counts are 40/40, 15/15 and 100/100; each session adds eight, three or 20 classes. Base pools follow long-tailed sampling. ImageNet-R uses task-specific splits after removing byte-identical duplicates. Novelty is relative

to task fitting, with CLIP's pre-training exposure retained. Detailed pools, preprocessing and seeds are in Supplementary Sections C, I and K.

Every comparator shares its encoder, partition and support. We compare fixed-gate classifier removal, independently recalibrated removal, dense anchored text–prototype fusion, summed/class-mean Tip-Adapter caches, and raw/fixed-temperature retained fusion. The latter uses a single base-validation scale per branch. HyCal uses the author's heterogeneous cosine/Mahalanobis metrics, count weighting and Gaussian-CDF calibration, transferred to the same eight-prompt anchors and frozen encoders. Its covariance regularization and novel offset are selected by base-only pseudo-expansion. Supplementary Section M documents the pinned implementation, numerical fixtures and transfer settings. All mixtures use base-only selection; FeNeC and earlier three-partition auxiliary studies are retained in Supplementary Sections A–H.

Orders are averaged within a partition. Means and standard deviations describe ten class assignments on fixed images. Paired differences use identical cells; constrained comparisons use jointly feasible sets. Conditional paired 95% bootstrap intervals resample partition means 20,000 times. Average incremental accuracy is the mean macro accuracy over five arrivals. These intervals characterize class-assignment variation, conditional on the fixed data and encoders.

4.2. Novel recognition at a shared base requirement

At the descriptive 95% retention budget, the ImageNet-R L/14 table reaches 84.31% novel accuracy at 86.42% base accuracy. Figure 2 compares novel recognition under shared base requirements across encoders and datasets. A more discriminative base readout can tolerate greater novel competition

while meeting the same target, turning retained discrimination into admission capacity.

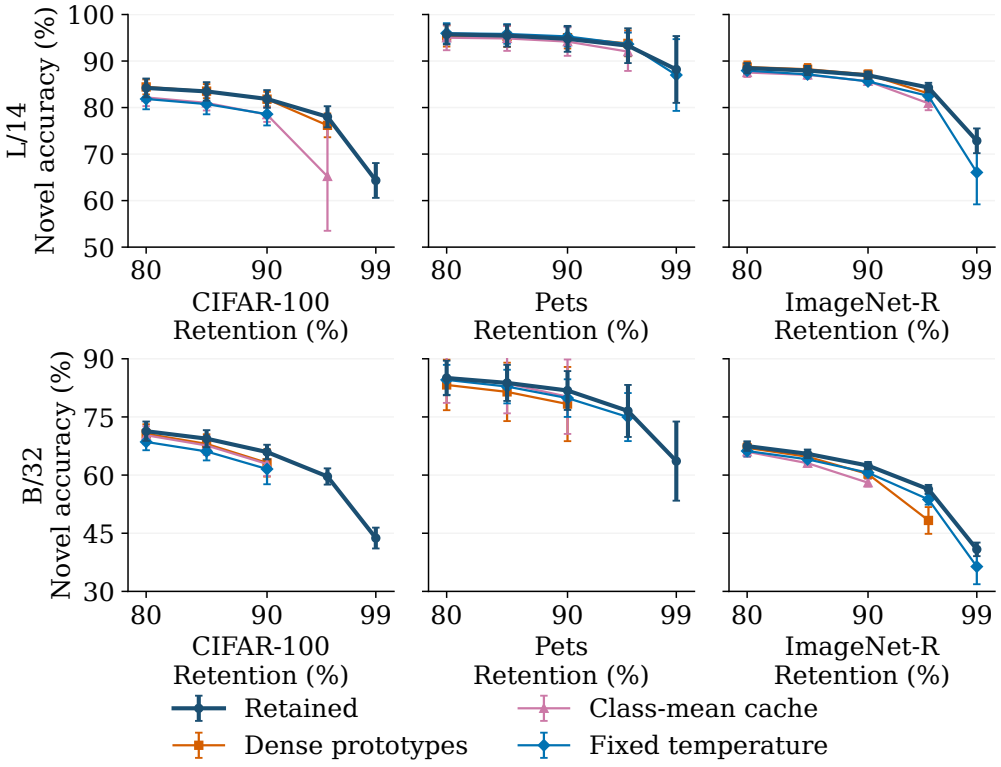


Figure 2: Novel recognition at shared base-retention requirements after the fifth arrival. Each readout must attain the stated fraction of the retained reference's closed-base accuracy within the same partition and encoder. Means and sample standard deviations cover ten partitions after averaging two arrival orders; points require all ten partitions to be feasible. Vertical ranges are 50–100% (L/14) and 30–90% (B/32); error bars stop at panel limits. Test-label threshold sweeps characterize descriptive capacity. Partial feasible sets and exact counts are reported in Supplementary Section I.

4.3. Query-adaptive alignment expands retention frontiers

Independent fusion tuning isolates query normalization under equal search budgets, with encoder, ridge fit, support and data roles fixed. At the common 95% base target, query normalization exceeds uncentered fusion by 3.60 [2.55,4.74] points on ImageNet-R L/14 and 5.29 [3.74,7.25] on B/32. Gains over fixed global normalization are 2.84 [2.42,3.28] and 4.65 [3.57,5.71] points. Each comparison includes ten partitions after averaging orders. Supplementary Section Q gives the factorial and search specification; Section M separately isolates centering and scaling with fusion held fixed.

The adaptive offset acts through the combined group margin in Eq. (7). On the original frozen models, adding its centering term raises base/novel ranking AUROC from .924 to .977 on ImageNet-R L/14 and from .856 to .904 on B/32. The term alone has novel-positive AUROC below .5 in all six conditions: its role is to align evidence within the decision margin. Section P supplies the conditional distributions and offset-reselected controls.

Independent tuning also gives a retained-minus-dense capacity gap of 9.69 [8.08,11.21] points on ImageNet-R B/32 and 0.25 [−0.46,0.92] on L/14. The next analyses examine how these recognition structures support repeated and certified admission.

4.4. The classifier's value across arrivals

Table 1 quantifies retained-minus-comparator average incremental accuracy with fixed base-only selected gates and offsets. On ImageNet-R, retaining the classifier adds 0.71 [0.46, 0.92] points over dense prototypes with L/14 and 0.90 [0.69, 1.11] with B/32. Corresponding gains over class-mean caches are

1.42 [1.15, 1.62] and 1.71 [1.44, 1.98] points. The result persists across both encoders and after calibration is restored to the classifier-free alternative.

Table 1: The retained classifier's value across five arrivals. Cells show paired gains in average incremental accuracy (percentage points), with conditional paired 95% bootstrap intervals below. Positive values favor retention; bold estimates have intervals excluding zero. Two orders are averaged within each of ten partitions. Full final-session differences are in Supplementary Section I.

Comparator	Retained-minus-comparator gain (pp) ↑		
	CIFAR-100	Pets	ImageNet-R
CLIP ViT-L/14			
Recalibrated removal	**+3.00** [+2.23, +3.75]	-0.14 [-1.53, +0.86]	**+2.30** [+1.95, +2.59]
Dense prototypes	+0.15 [-0.46, +0.64]	-0.98 [-2.52, +0.22]	**+0.71** [+0.46, +0.92]
Class-mean cache	**+1.44** [+0.68, +2.10]	-0.06 [-1.76, +1.41]	**+1.42** [+1.15, +1.62]
CLIP ViT-B/32			
Recalibrated removal	**+2.38** [+1.44, +3.34]	-0.44 [-1.64, +0.81]	**+3.07** [+2.82, +3.34]
Dense prototypes	+0.86 [-0.15, +1.76]	-0.39 [-1.85, +1.02]	**+0.90** [+0.69, +1.11]
Class-mean cache	+0.96 [-0.04, +1.84]	-0.74 [-1.62, +0.12]	**+1.71** [+1.44, +1.98]

CIFAR also shows gains over recalibrated removal with both encoders; its dense-prototype intervals span zero. Pets' average incremental intervals span zero for these comparators. The gain over prototype replacement is clearest on ImageNet-R.

Supplementary Section M relates closed-base discrimination to class fre-

quency and prototype quality and reports a controlled learned-branch intervention. It also evaluates the author-code-checked HyCal transfer: ImageNet-R average incremental accuracy is 76.36% with L/14 and 54.19% with B/32, with final 95% capacity feasible in two and zero partitions when both orders must qualify. These results describe the matched transfer under the shared retention requirement.

4.5. Class coverage enables certified admission

Independent calibration separates fitting, base-only tuning and class-stratified certification. The readouts and threshold candidates are frozen before varying calibration coverage; all methods receive the same images. Nested per-class caps of 2, 5, 10, 20 and 40 evaluate the label budget, using $r = 0.95$, $\delta = 0.05$ and five arrivals. Exact data roles, subset aggregation and the development history of the source benchmarks are specified in Supplementary Sections J and L.

Equation (13) requires effective coverage of at least 89.78 for the protection bound. The original CIFAR and Pets allocations provide 80 and 45–60, respectively. Figure 3 shows how increasing class coverage enables finite admission. At cap 40, the required base-label counts are 1,600 for CIFAR, 600 for Pets and 3,636–3,778 for ImageNet-R. These calibration labels are additional to five-shot novel support and can come from a reserved labeled catalog audit pool.

At cap 40, retained readouts certify final-session expansion in all ten partitions of each dataset and encoder, with all test targets met (Table 2). L/14 novel accuracy is 72.76% on CIFAR, 88.51% on Pets and 80.74% on ImageNet-R. All 20 arrival paths per dataset/encoder meet every test target.

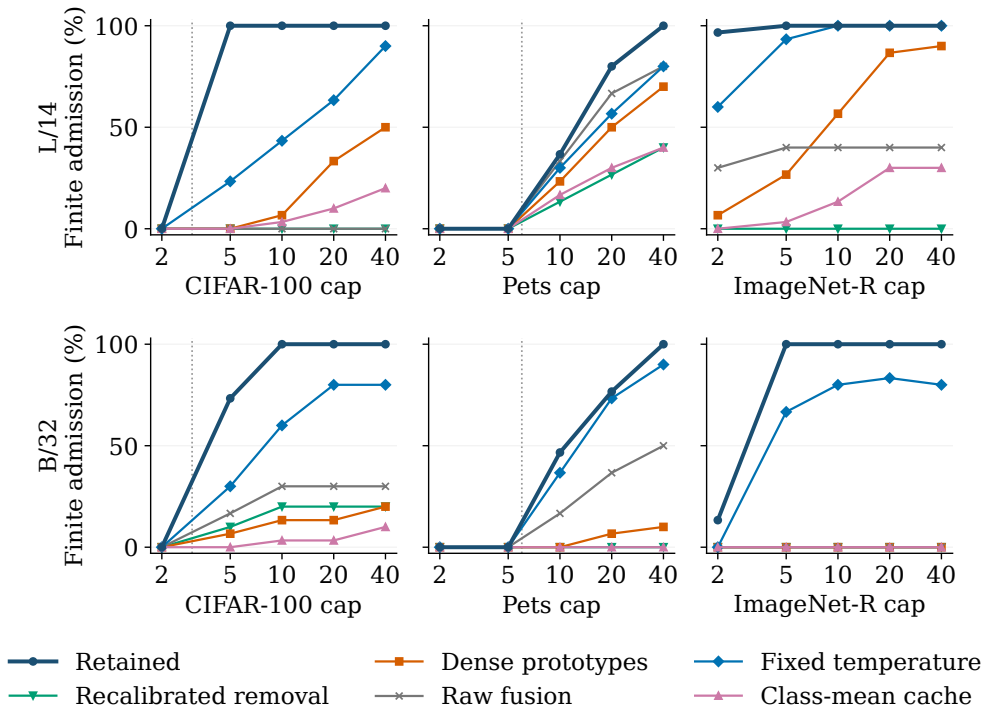


Figure 3: Finite certified admission versus calibration coverage, fifth arrival in the first order. All readouts share images and the protection rule. Grey lines show the necessary balanced cap from Eq. (13); ImageNet-R's floor lies below the plotted range. Below cap 40, three subset seeds are averaged within each of ten partitions; cap 40 uses the full pool once. Fallback counts as no finite admission. Novel-accuracy curves and exact counts appear in Supplementary Section L.

Finite admission holds throughout all CIFAR and ImageNet-R paths and 17/20 L/14 and 16/20 B/32 Pets paths; the remaining Pets paths use an early-session fallback. Full comparisons distinguish policy gains, which include fallback, from accuracy differences on jointly admitted partitions.

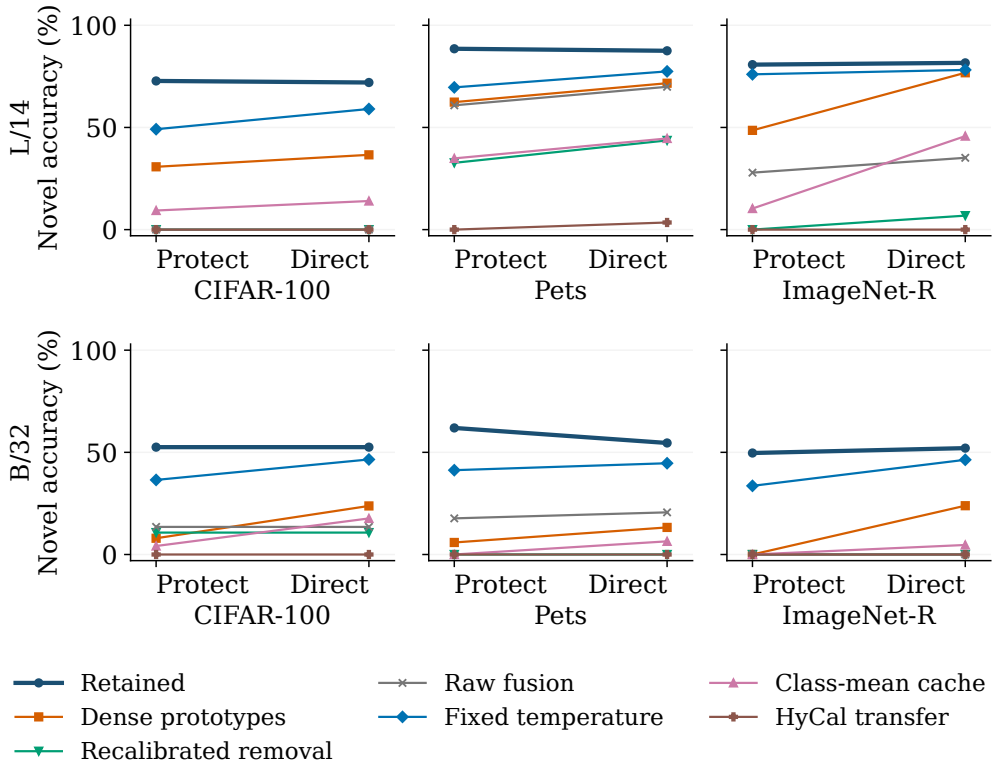


Figure 4: Calibration-rule comparison at identical cap-40 coverage, fifth arrival, first prespecified order. Lines connect the sufficient protection test and the direct paired accuracy test for each readout. Novel means cover all ten partitions, with zero for reference fallback. Each test uses 95% retention and 95% confidence over five prespecified arrivals within a fixed configuration. Admission counts, conditional accuracy and empirical target attainment are reported in Supplementary Section N.

4.6. From recognition capacity to certified policies

Figure 4 holds calibration images, candidate thresholds and error allocation fixed while changing the statistical criterion. On ImageNet-R L/14, crediting recovered errors raises dense-prototype admission from 9/10 to 10/10 and novel accuracy from 48.58% to 76.76%, including fallback. Retained readouts

Table 2: Protection-risk-calibrated admission after the fifth arrival in the first prespecified order, with up to 40 images per base class, 95% retention and 95% five-session confidence. Queries are actual base-calibration counts; Finite counts certified expansion and Test met counts attained test targets. Base/Novel are test percentages, mean ± sample standard deviation over ten partitions, including fallback. All six readouts are reported in Supplementary Section L.

Dataset	Calibration	Admission		Test accuracy (%) ↑	
	Queries	Finite	Test met	Base	Novel
CLIP ViT-L/14					
CIFAR-100	1600	10/10	10/10	85.86 ± 1.19	72.76 ± 4.76
Pets	600	10/10	10/10	94.10 ± 2.27	88.51 ± 4.85
ImageNet-R	3636–3778	10/10	10/10	89.02 ± 0.51	80.74 ± 1.83
CLIP ViT-B/32					
CIFAR-100	1600	10/10	10/10	76.74 ± 1.73	52.54 ± 4.34
Pets	600	10/10	10/10	88.10 ± 3.41	61.94 ± 14.07
ImageNet-R	3636–3778	10/10	10/10	72.98 ± 1.21	49.67 ± 2.38

attain 81.61%, a paired gain of 4.85 [2.78,6.93] points with all ten partitions admitted by both methods.

Let $F_m(t)$ be readout m's empirical novel capacity at base requirement t, N_m its selected-policy novel accuracy, and $H_m(t) = F_m(t) - N_m$ its remaining capacity. For retained R and dense P readouts,

$$N_R - N_P = \underbrace{F_R(t) - F_P(t)}_{\text{capacity difference}} + \underbrace{H_P(t) - H_R(t)}_{\text{remaining-capacity difference}} . \tag{16}$$

On ImageNet-R L/14, capacities 83.93% and 83.64% differ by 0.29 points,

while remaining capacities are 2.32 and 6.88 points. The latter difference accounts for 4.56 points of the 4.85-point policy gap. Comparable capacity can therefore coexist with different certified operating points. The remaining capacity combines threshold discretization, sampling and selection effects; this decomposition does not assign all of it to test variance.

Table 3 extends the analysis across datasets and encoders. The 35.40-point CIFAR L/14 policy gap includes reference fallback, alongside a 3.68-point capacity gap. Pets L/14 has a -2.12-point retained-minus-dense capacity gap and a positive policy gap. ImageNet-R B/32 additionally supplies a 10.40-point capacity advantage. Figure 5 overlays selected policies on exact empirical trajectories, displaying both capacity and how it is used.

Table 3: Retained-minus-dense selected-policy and descriptive capacity gains (pp), fifth arrival, first order. Capacity uses the common 95% minimum base target. Within a row, both columns use the same jointly feasible partitions (count); policy means include reference fallback. Brackets give conditional paired 95% bootstrap intervals on fixed images.

Encoder	Data	Policy gap	Common-floor capacity gap	Matched count
L/14	CIFAR-100	+35.40 [+16.61,+54.61]	+3.68 [+1.08,+6.38]	10/10
L/14	Pets	+15.88 [-1.27,+40.52]	-2.12 [-4.07,-0.49]	10/10
L/14	ImageNet-R	+4.85 [+2.77,+6.93]	+0.29 [-0.55,+1.16]	10/10
B/32	CIFAR-100	+26.38 [+11.60,+41.41]	+3.50 [-0.47,+7.94]	9/10
B/32	Pets	+42.92 [+15.13,+66.79]	+1.17 [-5.91,+7.38]	8/10
B/32	ImageNet-R	+26.15 [+15.85,+37.54]	+10.40 [+7.31,+14.64]	9/10

Matching at the dense policy's actual, stricter base accuracy gives a 3.23 [1.48,5.02]-point retained-minus-selected-dense gain on ImageNet-R L/14, with mean base overshoot 0.027 points. Comparing both frontiers at that

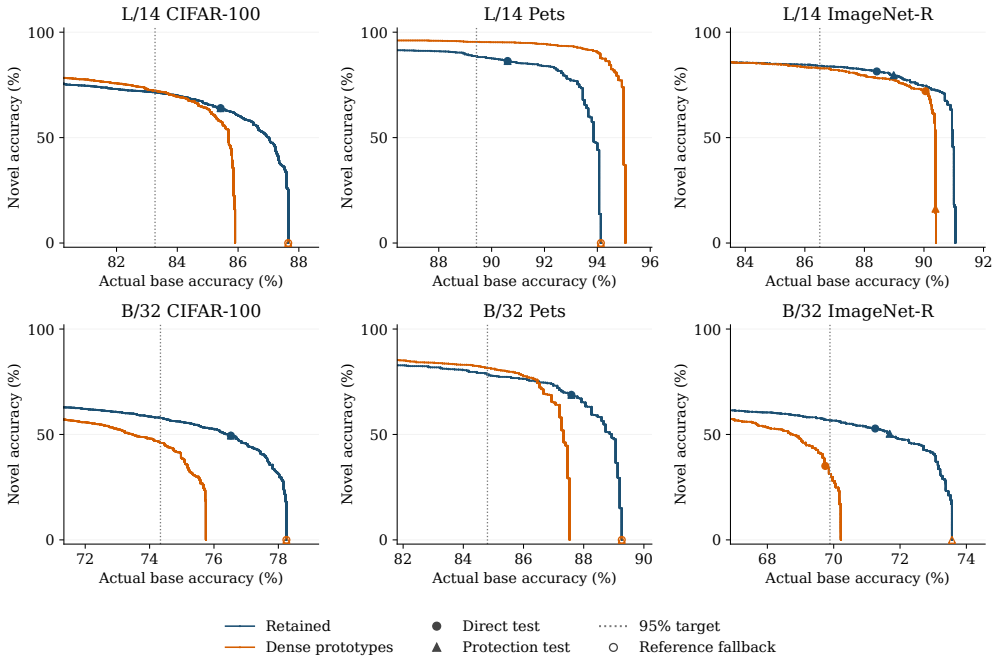


Figure 5: Capacity and certified operating points for prespecified partition 0, fifth arrival. The vertical line is the shared minimum base target. Triangles denote protection calibration, circles direct calibration, and hollow markers reference fallback. Curves use test labels to describe capacity; calibration chooses the points independently. Lines guide the eye; matching uses exact discrete events. Supplementary Section P provides all partitions and both orders.

stricter target gives 3.10 [1.37,4.88] points. Together, the common-floor and realized-accuracy views identify where the advantage lies. Full comparisons report joint feasibility, admission-conditioned accuracy and fallback, including the single dense B/32 ImageNet-R test-target miss (Sections N and P).

4.7. Target calibration transfers frozen source readouts

The controlled CIFAR stress test uses clean fitting/support and noisy or blurred queries. Clean calibration meets 5/10 and 1/10 L/14 final targets;

matching calibration restores 10/10 with finite admission for both encoders (Section O).

The separate transfer study freezes the original source-selected ImageNet-R readouts and evaluates matching ImageNet-Sketch classes [32]. Identity-disjoint target calibration and test pools are fixed before scoring (Supplementary Section R). Source and Sketch calibration use equal per-class counts; model parameters and source support remain fixed. With direct Sketch calibration, retained readouts reach 86.82% and 58.96% novel accuracy with L/14 and B/32, respectively, with finite admission and target attainment in all ten partitions. Source calibration also meets these targets, at 82.99% and 54.87% novel accuracy. At the common 95% minimum target, L/14 retained and dense capacities both round to 88.84%; B/32 retention supplies a 5.78 [4.03,7.55]-point capacity gain on nine jointly feasible partitions. Section R reports the identity exclusions, absolute base accuracies, both calibration criteria and exact frontiers.

5. Conclusion

Reference-anchored evidence alignment supports class expansion with finite-sample retention control. Fixed base references preserve existing discrimination as the vocabulary grows, while query normalization induces an adaptive admission margin. Preserved base rankings make protection coincide with the accuracy requirement; a recovery-aware test extends certification to replacement readouts. Controlled comparisons establish the normalization mechanism, and capacity-to-policy analysis shows how comparable recognition frontiers can yield different certified admission. Target calibration transfers

frozen source readouts to natural rendition shift without retraining. The framework connects stable evidence, explicit accuracy requirements and useful admission across successive class arrivals.

CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets, ImageNet-R and ImageNet-Sketch data and the publicly released CLIP ViT-L/14 and ViT-B/32 checkpoints. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-T`

ables. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

References

[1] Z. Liu, Z. Miao, X. Zhan, J. Wang, B. Gong, S. X. Yu, Large-Scale Long-Tailed Recognition in an Open World, arXiv preprint arXiv:1904.05160 (2019).
URL https://arxiv.org/abs/1904.05160

[2] K. Cao, C. Wei, A. Gaidon, N. Arechiga, T. Ma, Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss, arXiv preprint arXiv:1906.07413 (2019).
URL https://arxiv.org/abs/1906.07413

[3] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-Balanced Loss Based on Effective Number of Samples, arXiv preprint arXiv:1901.05555 (2019).
URL https://arxiv.org/abs/1901.05555

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning Transferable Visual Models From Natural Language Supervision, in: International Conference on Machine Learning, 2021, pp. 8748–8763.
URL https://proceedings.mlr.press/v139/radford21a.html

[5] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Learning to Prompt for Vision-Language Models, International Journal of Computer Vision 130 (9) (2022) 2337–2348. doi:10.1007/s11263-022-01653-1.
URL https://link.springer.com/article/10.1007/s11263-022-01653-1

[6] J.-X. Shi, T. Wei, Z. Zhou, J.-J. Shao, X.-Y. Han, Y.-F. Li, Long-Tail Learning with Foundation Model: Heavy Fine-Tuning Hurts, in: International Conference on Machine Learning, 2024, pp. 45014–45039.
URL https://proceedings.mlr.press/v235/shi24g.html

[7] J. Snell, K. Swersky, R. S. Zemel, Prototypical Networks for Few-shot Learning, arXiv preprint arXiv:1703.05175 (2017).
URL https://arxiv.org/abs/1703.05175

[8] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning, arXiv preprint arXiv:1911.04623 (2019).
URL https://arxiv.org/abs/1911.04623

[9] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification,

arXiv preprint arXiv:2207.09519 (2022).
URL https://arxiv.org/abs/2207.09519

[10] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, C. H. Lampert, iCaRL: Incremental Classifier and Representation Learning, arXiv preprint arXiv:1611.07725 (2016).
URL https://arxiv.org/abs/1611.07725

[11] X. Tao, X. Hong, X. Chang, S. Dong, X. Wei, Y. Gong, Few-Shot Class-Incremental Learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12183–12192.
URL https://openaccess.thecvf.com/content_CVPR_2020/html/Tao_Few-Shot_Class-Incremental_Learning_CVPR_2020_paper.html

[12] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, Z. Liu, Revisiting Class-Incremental Learning with Pre-Trained Models: Generalizability and Adaptivity are All You Need, International Journal of Computer Vision (2024).
URL https://github.com/LAMDA-CL/RevisitingCIL

[13] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, A. van den Hengel, RanPAC: Random Projections and Pre-trained Models for Continual Learning, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 12022–12053. doi:10.52202/075280-0526.
URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/2793dc35e14003dd367684d93d236847-Abstract-Conference.html

[14] X. Xu, S. Niu, Z. Wang, W. Guo, L. Jing, H. Yang, Multi-feature space similarity supplement for few-shot class incremental learning, Knowledge-Based Systems 265 (2023) 110394. doi:10.1016/j.knosys.2023.110394.

[15] R. Chen, X.-Y. Jing, F. Wu, H. Chen, Sharpness-aware gradient guidance for few-shot class-incremental learning, Knowledge-Based Systems 299 (2024) 112030. doi:10.1016/j.knosys.2024.112030.

[16] C. Han, Z. Gu, C. Xu, Z. Cui, Adaptive exploration for few-shot incremental learning, Knowledge-Based Systems 330 (2025) 114496. doi:10.1016/j.knosys.2025.114496.

[17] R. He, D. Fang, Y. Chen, K. Tong, C. Chen, Y. Wang, L.-p. Chau, H. Zhuang, REAL: Representation enhanced analytic learning for exemplar-free class-incremental learning, Knowledge-Based Systems 332 (2026) 114901. doi:10.1016/j.knosys.2025.114901.

[18] H. Fang, H. Cai, K. Mao, J. Zhao, X. Xiao, Few-shot class-incremental learning based on prompt guidance and multimodal fusion, Knowledge-Based Systems 342 (2026) 115862. doi:10.1016/j.knosys.2026.115862.

[19] L. Li, Z. Wu, Y. Ji, MoTE: Mixture of task-specific experts for pre-trained model-based Class-incremental learning, Knowledge-Based Systems 324 (2025) 113795. doi:10.1016/j.knosys.2025.113795.

[20] M. Dong, Z. Zhang, X. Tan, J. Qiu, Y. Xie, Dynamic expansion orthogo-

nal network for class-incremental learning, Knowledge-Based Systems 341 (2026) 115868. doi:10.1016/j.knosys.2026.115868.

[21] J. Yuan, W. Li, L. Li, L. Zhang, E. Ning, X. Gao, X. Ning, Counterfactual distribution intervention for few-shot class-incremental learning, Knowledge-Based Systems 351 (2026) 116754. doi:10.1016/j.knosys.2026.116754.

[22] Q. Yan, Y. Yang, Y. Dai, X. Zhang, K. Wiltos, M. Woźniak, W. Dong, Y. Zhang, CLIP-guided continual novel class discovery, Knowledge-Based Systems 310 (2025) 112920. doi:10.1016/j.knosys.2024.112920.

[23] K. Książek, H. Jastrzębski, K. Pniaczek, B. Trojan, M. Karp, J. Tabor, FeNeC: Enhancing continual learning via feature clustering with neighbor- or logit-based classification, Knowledge-Based Systems 338 (2026) 115479. doi:10.1016/j.knosys.2026.115479.

[24] Y. Chen, T. Ding, L. Wang, J. Huo, Y. Gao, W. Li, Enhancing few-shot class-incremental learning via training-free bi-level modality calibration, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 9881–9890.
URL https://openaccess.thecvf.com/content/CVPR2025/html/Chen_Enhancing_Few-Shot_Class-Incremental_Learning_via_Training-Free_Bi-Level_Modality_Calibration_CVPR_2025_paper.html

[25] E. Lee, M. Kim, J. Kwon, Y. Lee, J. Kim, S. Jang, Y. Kim, Hycal: A training-free prototype calibration method for cross-discipline few-shot class-incremental learning, in: Proceedings of the IEEE/CVF Conference

on Computer Vision and Pattern Recognition, 2026, pp. 29462–29471.
URL https://arxiv.org/abs/2604.15678

[26] W.-L. Chao, S. Changpinyo, B. Gong, F. Sha, An Empirical Study and Analysis of Generalized Zero-Shot Learning for Object Recognition in the Wild, arXiv preprint arXiv:1605.04253 (2016).
URL https://arxiv.org/abs/1605.04253

[27] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, arXiv preprint arXiv:2110.01052 (2021). doi:10.48550/arXiv.2110.01052.

[28] I. Waudby-Smith, A. Ramdas, Estimating means of bounded random variables by betting, Journal of the Royal Statistical Society Series B: Statistical Methodology 86 (1) (2024) 1–27. doi:10.1093/jrsssb/qkad009.

[29] A. Krizhevsky, Learning multiple layers of features from tiny images, Tech. rep., University of Toronto (2009).
URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[30] O. M. Parkhi, A. Vedaldi, A. Zisserman, C. V. Jawahar, Cats and dogs, in: IEEE Conference on Computer Vision and Pattern Recognition, 2012, pp. 3498–3505. doi:10.1109/CVPR.2012.6248092.
URL https://www.robots.ox.ac.uk/~vgg/publications/2012/parkhi12a/

[31] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, J. Gilmer, The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution Generalization, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8340–8349.
URL https://openaccess.thecvf.com/content/ICCV2021/html/Hendrycks_The_Many_Faces_of_Robustness_A_Critical_Analysis_of_Out-of-Distribution_ICCV_2021_paper.html

[32] H. Wang, S. Ge, Z. C. Lipton, E. P. Xing, Learning robust global representations by penalizing local predictive power, in: Advances in Neural Information Processing Systems, Vol. 32, 2019, pp. 10506–10518.